Supplementary Materials:
mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA

S1. Hyperparameters

S1.1 Pre-training

Table 1: Pre-training hyperparameters for Phase 1 and Phase 2.

Hyperparameter	**Phase 1**	**Phase 2**
Training steps	50,000	25,000
Warmup steps	2,000	500
Peak learning rate	1×10^{-4}	3×10^{-5}
LR schedule	Cosine decay	Cosine decay
Batch size (effective)	128	128
Gradient accumulation steps	8	8
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)	AdamW
Weight decay	0.01	0.01
Gradient clipping	1.0	1.0
MLM masking probability	0.15	0.15
Heteroplasmy loss weight (λ_{het})	0.0	0.3
Checkpoint rotation	Keep last 3	Keep last 3
Gradient checkpointing	Yes	Yes

S1.2 Model Architecture

S2. Mathematical Derivation: Circular Positional Encoding

S2.1 Standard sinusoidal PE

In the original Transformer [Vaswani et al., 2017], position p is encoded as:

$$\mathrm{PE}(p, 2i) = \sin\Big(\frac{p}{10000^{2i/d}}\Big) \tag{1}$$

$$\mathrm{PE}(p, 2i+1) = \cos\Big(\frac{p}{10000^{2i/d}}\Big) \tag{2}$$

For a sequence of length L, $\mathrm{PE}(0, \cdot) \neq \mathrm{PE}(L-1, \cdot)$ in general. As $d \to \infty$, the angle between PE(0) and PE($L-1$) grows monotonically with L.

Table 2: mtDNA-FM architecture hyperparameters.

Parameter	**Value**
Hidden size (d)	256
Number of transformer layers	6
Number of attention heads	8
Feed-forward intermediate size	1,024
Attention head dimension	32
Dropout	0.1
Attention dropout	0.1
Max position embeddings	514 (512 + [CLS] + [SEP])
Vocabulary size	4,102
Activation function	GELU
Layer normalization ϵ	10^{-12}
Total parameters	≈5.8M
Positional encoding	Circular sinusoidal (fixed buffer)
Genome length for circular PE	16,569 bp (rCRS)

S2.2 Circular modification

We replace p with the circular angle $\phi_p = 2\pi p / L_{\text{genome}}$:

$$\text{PE}_{\text{circ}}(p, 2i) = \sin\left(\frac{2\pi p / L_{\text{genome}}}{10000^{2i/d}}\right) \tag{3}$$

$$\text{PE}_{\text{circ}}(p, 2i+1) = \cos\left(\frac{2\pi p / L_{\text{genome}}}{10000^{2i/d}}\right) \tag{4}$$

At $p = L_{\text{genome}}$:

$$\phi_{L_{\text{genome}}} = \frac{2\pi L_{\text{genome}}}{L_{\text{genome}}} = 2\pi$$

and $\sin(2\pi) = \sin(0) = 0$, $\cos(2\pi) = \cos(0) = 1$.

Therefore $\text{PE}_{\text{circ}}(L_{\text{genome}}, \cdot) = \text{PE}_{\text{circ}}(0, \cdot)$, which is the desired circular continuity property.

S2.3 Properties

- *Uniqueness:* Positions p and p' with $|p - p'| < L_{\text{genome}}/2$ receive distinct encodings (the mapping $p \mapsto \phi_p$ is injective on $[0, L_{\text{genome}})$).
- *Smoothness at junction:* $\|\text{PE}_{\text{circ}}(p) - \text{PE}_{\text{circ}}(p+1)\|$ is constant across all positions including $p = L_{\text{genome}} - 1$ (the circular junction), whereas for standard sinusoidal PE this distance jumps at position 0.
- *Non-learnable:* Implemented as `nn.Buffer`, not `nn.Parameter`; gradients do not flow through the positional encoding.

S3. Per-class Zero-shot Haplogroup Metrics

S3.1 Zero-shot 5-NN (26-class evaluation)

Per-class precision, recall, and F1 from zero-shot cosine 5-nearest-neighbour classification on 757 test sequences across 26 major PhyloTree haplogroup classes. No haplogroup labels were used in pre-training. Sequences from NCBI haplogroup-labeled human mitochondrial genomes: 15,741 total records were retrieved; 15,364 had a non-null haplogroup field; 13,884 could be assigned to one of the 26 major PhyloTree classes and were used for evaluation (stratified 80/10/10 split, `random_state=42`).

Table 3: Per-class zero-shot 5-NN haplogroup metrics (757 test queries, 1,509 train library sequences, cosine distance). No haplogroup labels used in pre-training. Classes with $N < 10$ test sequences (E=4, L5=4, G=10, L4=11, V=11) have high per-class metric variance; interpret their F1 scores with caution.

Haplogroup	**Precision**	**Recall**	**F1**	**N test**
A	0.218	0.475	0.299	40
B	0.667	0.800	0.727	40
C	0.667	0.600	0.632	40
D	0.068	0.100	0.081	40
E	0.273	0.750	0.400	4
F	0.774	0.667	0.716	36
G	0.098	0.400	0.157	10
H	0.061	0.050	0.055	40
HV	0.036	0.040	0.038	25
I	0.762	0.800	0.780	40
J	0.185	0.125	0.149	40
K	0.182	0.150	0.164	40
L0	0.857	0.750	0.800	40
L1	0.727	0.571	0.640	28
L2	0.861	0.775	0.816	40
L3	0.485	0.400	0.438	40
L4	0.474	0.818	0.600	11
L5	1.000	0.750	0.857	4
M	0.412	0.175	0.246	40
N	0.133	0.095	0.111	21
R	0.857	0.316	0.462	19
T	0.133	0.050	0.073	40
U	0.389	0.175	0.241	40
V	0.056	0.091	0.069	11
W	0.000	0.000	0.000	16
X	0.071	0.083	0.077	12
Macro avg	**0.3703**			757

S3.2 Baseline: DNABERT-2 Zero-shot Haplogroup Classification

DNABERT-2 [Zhou et al., 2024] (117M parameters, BPE tokenization, pre-trained on 32 vertebrate nuclear genomes) was evaluated with the same cosine 5-NN protocol on the same train/test split as mtDNA-FM. Each 16,569 bp sequence was truncated to at most 3,000 nucleotides before tokenization; DNABERT-2's 512-token context limit applies after BPE tokenization. The CLS embedding from the final transformer layer was used as the sequence representation, matching the protocol for mtDNA-FM.

S4. Dataset Summary

S5. DVC Pipeline

The complete analysis pipeline is managed with DVC. The following stages are defined:

1. `download_hmtdb`: Download and cache HmtDB sequences
2. `download_ncbi`: Download vertebrate mtDNA from NCBI Entrez
3. `download_variants`: Download gnomAD, ClinVar, PhyloTree variant data
4. `preprocess`: Quality filter, normalize to rCRS length, train/val/test split

Table 4: Zero-shot 26-class haplogroup classification: DNABERT-2 vs mtDNA-FM. Both use cosine 5-NN on the same 1,509-sequence train library and 757-sequence test set. DNABERT-2's 512-token BPE context covers approximately 3,000 nt (18%) of each 16,569 bp genome; mtDNA-FM encodes the full genome via overlapping 512-token windows.

Model	Params	Accuracy (%)	Macro-F1
DNABERT-2 (nuclear-genome, 18% of mtDNA seen)	117M	66.3	0.6593
mtDNA-FM (mtDNA-specialized, full genome)	5.8M	**37.9**	0.3703

Table 5: Datasets used in this study.

Dataset	Source	Size	Purpose
HmtDB	Clima et al. [2017]	34,975 seqs used (47,000 total in database)	Phase 2 pre-train
NCBI vertebrate mtDNA	NCBI Entrez	117,615 seqs	Phase 1 pre-train
PhyloTree	van Oven and Kayser [2009]	26 major groups	Haplogroup labels
gnomAD v3.1 chrM	Karczewski et al. [2020]	419 benign SNPs (AF >1%)	Benign variants
ClinVar	Landrum et al. [2018]	118 pathogenic SNPs	Pathogenic variant

5. `build_vocabulary`: Construct 4,102-token k-mer vocabulary
6. `pretrain_phase1`: Phase 1 cross-species pre-training (50k steps)
7. `pretrain_phase2`: Phase 2 human specialization (25k steps)
8. `evaluate`: Compute all metrics, generate evaluation reports

Reproduce the full pipeline: `dvc repro`

S6. Ablation Learning Curves

Supplementary Table S3.2 reports the DNABERT-2 zero-shot baseline comparison (same k-NN protocol, same train/test split). Architectural ablations (circular PE vs. sinusoidal PE; with vs. without heteroplasmy channel) and the Phase 1 vs Phase 2 curriculum ablation require GPU compute and are deferred to the extended journal paper.

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